

CS505 Project Status Report: Citation-Grounded QA with Local Models

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1 Project Goal

The project studies hallucination reduction in scientific question answering using a fully local retrieval-augmented generation pipeline. The current stack is built around QASPER, shared SCC storage, and locally-run small instruction-tuned models. The main experimental question is how much answer quality and hallucination behavior depend on retrieval quality versus generation behavior, especially under citation forcing.

2 SCC Layout and Reproducibility

The implementation uses a strict split between shared deterministic artifacts and per-user experimental outputs.

Shared course path

```
/projectnb/cs505am/projects/zukhriddin_nlp_research_project/qasper
```

Student working repo

```
/projectnb/cs505am/students/zfmsai/zukhriddin_nlp_research_project
```

Backward-compatible symlink

```
/projectnb/cs505am/students/zfmsai/nlp_research_project
```

The code automatically resolves either the renamed project path or the older path. Shared data and processed artifacts live in `projects/`; model caches, runs, and temporary indexes live in `students/$USER/`.

3 What Was Implemented

3.1 Task 2: Data and Chunking

Implemented components:

- QASPER loader from official JSON/Hugging Face format
- Internal normalized schema for papers, questions, answers, and chunks
- Fixed chunking
- Sliding-window chunking
- Section-aware chunking
- Deterministic processed split builder for team-shared reuse

Canonical shared processed artifacts were built for:

- `train`
- `validation`
- `test`

These outputs are the interface boundary used by both retrieval and generation.

3.2 Task 3: Retrieval

Implemented retrieval stack:

- BM25
- Dense retrieval using `BAAI/bge-small-en-v1.5`
- Hybrid reciprocal-rank fusion
- Cross-encoder reranking using `cross-encoder/ms-marco-MiniLM-L-6-v2`
- Evaluation via Recall@5, MRR@10, and evidence hit rate

3.3 Task 4: Generation and Evaluation

Implemented generation stack:

- Baseline prompt style
- Citation-forcing prompt style
- Local generation with `microsoft/Phi-3.5-mini-instruct` via `transformers`
- Answer normalization for canonical QASPER answer forms (yes, no, unanswerable)
- Context compression to question-relevant sentences
- SCC-safe batch scripts and memory settings
- Evaluation via token F1, citation precision, citation rate, hallucination proxy, and supported-answer rate

3.4 Task 1: Infra and SCC

Implemented infrastructure:

- Reproducible path/config system
- SCC batch scripts for retrieval and generation
- Student-local Hugging Face cache handling
- Documentation for SCC environment setup
- GPU-safe defaults that run on `academic-gpu`

4 How the Pipeline Works

1. Load QASPER and normalize into internal objects.
2. Build processed split artifacts shared across the team.
3. Chunk each paper with one of the chunking strategies.
4. Retrieve from chunks belonging only to the paper for the given question.
5. Generate an answer from the top retrieved chunks.
6. Score the generated answer against gold answers and gold evidence.

The current best retriever for generation is `hybrid`; `hybrid_rerank` was evaluated later as a stronger comparator.

5 Main Results

5.1 Retrieval Results on Validation (Section Chunks)

Method	Recall@5	MRR@10	Evidence Hit
BM25	0.5769	0.4110	0.7152
Dense	0.5886	0.4820	0.7238
Hybrid	0.6313	0.5097	0.7778
Hybrid + rerank	0.6410	0.5027	0.7702

Table 1: Validation retrieval results on section-aware chunks.

Takeaway: Hybrid retrieval is the strongest overall default for generation; hybrid + rerank improves recall slightly but did not clearly dominate all retrieval metrics.

5.2 Final Validation Results for the Core Phi Conditions

Condition	Questions	Token F1	BERTScore	Citation Prec.	Halluc.	Supported
A fixed + dense + baseline	1005	0.3059	0.8791	0.0000	0.5134	0.6886
B fixed + hybrid + baseline	1005	0.3063	0.8782	0.0000	0.5104	0.6716
P parametric-only	1005	0.1350	0.8329	0.0000	0.8657	0.1343
M BM25-only	1005	0.3207	0.8813	0.0000	0.4716	0.7254
C section + hybrid + baseline	1005	0.3197	0.8815	0.0000	0.4527	0.7741
D section + hybrid_rerank + baseline	1005	0.3238	0.8817	0.0000	0.4478	0.7652
E section + hybrid_rerank + citation	1005	0.3215	0.8819	0.3230	0.5463	0.5960

Table 2: Full-validation generation results on 1005 validation questions. Condition E also achieved citation rate 0.9274 and NLI citation precision 0.1416.

Takeaway: The full A–E matrix is now on the same full-validation footing. Fixed-chunk conditions A and B underperform the section-aware conditions, which strengthens the original proposal hypothesis that better chunk structure materially helps answer quality. The best overall full-validation setting remains D: section-aware chunking, hybrid retrieval, reranking, and baseline prompting. Citation forcing (E) produces citations very reliably, but still lowers grounding and answer support relative to the strongest baseline setting.

5.3 Post-Full-Validation Citation Tuning

After the full-validation comparison, several targeted citation-only fixes were tested on a 100-question validation slice. The goal was to reduce unsupported cited claims without changing the retrieval setup.

Run	Token F1	Citation Prec.	Halluc.	Supported
Earlier citation sample (overlap baseline)	0.3569	0.3528	0.5100	0.6900
Strict one-sentence contract	0.2976	0.3755	0.5200	0.6700
Decimal-safe compactor + relaxed prompt	0.3234	0.3032	0.5900	0.5900
Vacuous-restatement gate	0.3288	0.3032	0.5800	0.6100

Table 3: Citation-only tuning runs on the same 100-question overlap.

Takeaway: deterministic post-processing recovered some answer quality after the strict-prompt regression, but none of the new citation variants beat the earlier stronger citation sample on the same 100-question overlap. As a result, citation tuning is paused and D remains the preferred overall system configuration.

5.4 Error Analysis on the Strongest Baseline and Citation Runs

- **Baseline (D):** retrieval misses remain the dominant failure mode (291 questions), followed by overcautious `unanswerable` predictions (144 questions). We also see a smaller but important lexical-mismatch slice (49 questions), where token F1 is low but BERTScore is high.
- **Citation forcing (E):** the same retrieval misses remain, but citation-specific issues dominate after retrieval succeeds. The error analysis found 790 unsupported-citation cases, 84 overcautious `unanswerable` cases, and 162 grounding failures despite evidence being retrieved.
- **Interpretation:** the current citation prompt encourages the model to cite frequently, but many cited claims are not actually entailed by the cited chunk. This matches the low NLI citation precision on full validation.

6 What Changed During Debugging

Several implementation issues were identified and fixed during the experiments:

- SCC scheduler instability prevented job submission for a period; jobs were moved to proper batch

usage once SCC recovered.

- The correct GPU queue for this course environment was `academic-gpu`.
- Generation initially hit CUDA OOM; the fix was to cap input/context lengths and use conservative CUDA settings.
- Token F1 was artificially low because the model often produced verbose forms like “The context does not specify...” instead of exact canonical answers. Answer normalization fixed this.
- SCC environment-variable submission did not reliably propagate `QUESTION_LIMIT`; the batch script was changed to accept it as a positional argument.
- After the SCC project rename, shared path resolution failed; config detection was extended to support both old and new project names.

7 Current Status

Completed:

- Task 2: complete
- Task 3: complete
- Task 4 core Phi-based evaluation pipeline: complete through full validation
- NLI citation backfill and BERTScore backfill: complete for the core Phi runs
- Task 1 setup, SCC usage, and reproducibility documentation: complete for the current stack
- Human-evaluation sample and reviewer templates: prepared and ready for annotation

Reached goals:

- deterministic shared QASPER preprocessing for the whole team
- full retrieval implementation and benchmarking
- full generation implementation and benchmarking with Phi-3.5
- SCC-compatible batch execution for both retrieval and generation
- clear best current setting before moving on to new models

Best current setting before Qwen:

- retrieval/generation: `hybrid_rerank` with baseline prompting is the strongest full-validation Phi configuration
- citation behavior: citation forcing is useful for explicit attribution, but not yet for best grounded answer quality

8 How to Inspect Results

From the local archive:

```
cd /Users/zukhriddin/Downloads/CS505_NLP/NLP_research_project
ls -lt local_scc_results/latest/runs/generation/validation/section/*.json
```

Human-evaluation materials are in:

```
reports/artifacts/human_eval_sample.csv
reports/artifacts/human_eval_template_reviewer_1.csv
reports/artifacts/human_eval_template_reviewer_2.csv
reports/artifacts/human_eval_instructions.md
```

To print metrics from a saved run:

```
python3 - <<'PY'
import json
from pathlib import Path
run_dir = Path("local_scc_results/latest/runs/generation/validation/section")
```

```

path = run_dir / "hybrid_rerank.baseline.20260413T002827Z.json"
obj = json.loads(path.read_text())
print(obj["metrics"])
PY

```

9 How to Reproduce the Pipeline

9.1 Environment

```

chmod +x setup_env.sh
./setup_env.sh

```

9.2 Check resolved paths

```
python3 scripts/show_project_paths.py
```

9.3 Retrieval evaluation

```

python scripts/evaluate_qasper_retrieval.py \
    --split validation \
    --strategy section \
    --method hybrid

```

9.4 Generation evaluation

```

python scripts/evaluate_qasper_generation.py \
    --split validation \
    --strategy section \
    --retrieval-method hybrid \
    --prompt-style baseline \
    --generation-model microsoft/Phi-3.5-mini-instruct

```

9.5 SCC batch generation sample

```

qsub -N phi_full_validation -o logs/phi_full_validation.log \
    scripts/submit_full_validation_phi.sh

```

9.6 Local SCC archive refresh

- Pull the latest SCC runs/ and logs/ down locally without deleting anything remotely:

```
bash scripts/pull_scc_results.sh
```

10 Next Step

The next remaining steps are evaluation and writeup work, not more infrastructure:

- fold the full-validation Phi results into the final report tables
- use the new error-analysis artifacts to write a short qualitative discussion section
- decide whether stretch-model experiments (Qwen, Llama, Mistral) are worth keeping as side results in the final report
- if needed, add one more citation-prompt iteration or a local-judge faithfulness metric